

Section 4 - Practice Problems

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Chapter 1: Limits of Functions

Exercise 4.1.4

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and let $c \in \mathbb{R}$. Show that $\lim_{x \rightarrow c} f(x) = L$ if and only if $\lim_{x \rightarrow 0} f(x+c) = L$.

Solution:

(I) Forward Direction.

$$\begin{aligned} |x - c| < \delta &\implies \\ |f(x) - L| &< \epsilon \\ |x| < \delta &\implies \\ |f(x+c) - L| &< \epsilon \end{aligned}$$

(II) Reverse Direction.

Exercise 4.1.5

Let $I := (0, a)$ where $a > 0$, and let $g(x) := x^2$ for $x \in I$. For any points $x, c \in I$, show that $|g(x) - c^2| \leq 2a|x - c|$. Use this inequality to prove that $\lim_{x \rightarrow c} x^2 = c^2$ for any $c \in I$.

Solution:**Exercise 4.1.8**

Show that $\lim_{x \rightarrow c} \sqrt{x} = \sqrt{c}$ for any $c > 0$.

Solution:**Exercise 4.1.12**

Show that the following limits do *not* exist.

(a) $\lim_{x \rightarrow 0} \frac{1}{x^2} \ (x > 0)$.

Solution:

(b) $\lim_{x \rightarrow 0} \frac{1}{\sqrt{x}} \ (x > 0).$

Solution:

(c) $\lim_{x \rightarrow 0} (x + \text{sign}(x)).$ Note that $\text{sign}(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -1, & \text{if } x < 0 \end{cases}.$

Solution:

(d) $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x^2}\right).$

Solution:

Exercise 4.1.14

Let $c \in \mathbb{R}$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that $\lim_{x \rightarrow c} (f(x))^2 = L.$

(a) Show that if $L = 0$, then $\lim_{x \rightarrow c} f(x) = 0.$

Solution:

(b) Show by example that if $L \neq 0$, then f may not have a limit at $c.$

Solution:

Chapter 2: Limit Theorems

Exercise 4.2.4

Prove that $\lim_{x \rightarrow 0} \cos\left(\frac{1}{x}\right)$ does not exist but that $\lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right) = 0$.

Solution:

Exercise 4.2.5

Let f, g be defined on $A \subseteq \mathbb{R}$ to $\mathbb{R}$, and let c be a cluster point of A . Suppose that f is bounded on a neighbourhood of c and that $\lim_{x \rightarrow c} g = 0$. Prove that $\lim_{x \rightarrow c} f \cdot g = 0$.

Solution:

Exercise 4.2.8

Let $n \in \mathbb{N}$ be such that $n \geq 3$. Derive the inequality $-x^2 \leq x^n \leq x^2$ for $-1 < x < 1$. Then use the fact that $\lim_{x \rightarrow 0} x^2 = 0$ to show that $\lim_{x \rightarrow 0} x^n = 0$.

Solution:

Exercise 4.2.9

Let f, g be defined on A to $\mathbb{R}$ and let c be a cluster point of A .

(a) Show that if both $\lim_{x \rightarrow c} f$ and $\lim_{x \rightarrow c} (f + g)$ exists, then $\lim_{x \rightarrow c} g$ exists.

Solution:

(b) If $\lim_{x \rightarrow c} f$ and $\lim_{x \rightarrow c} f \cdot g$ exist, does it follow that $\lim_{x \rightarrow c} g$ exists?

Solution:

No

Exercise 4.2.11

Determine whether the following limits exist in $\mathbb{R}$.

(a) $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x^2}\right) \quad (x \neq 0).$

Solution:

(b) $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x^2}\right) \quad (x \neq 0).$

Solution:

(c) $\lim_{x \rightarrow 0} \operatorname{sign}\left(\sin\left(\frac{1}{x^2}\right)\right) \quad (x \neq 0).$

Solution:

(d) $\lim_{x \rightarrow 0} \sqrt{x} \sin\left(\frac{1}{x^2}\right).$

Solution:

Exercise 4.2.15

Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$, and let $c \in \mathbb{R}$ be a cluster point of A . In addition, suppose that $f(x) \geq 0$ for all $x \in A$, and let $\sqrt{f}$ be the function defined for $x \in A$ by $(\sqrt{f})(x) := \sqrt{f(x)}$. If $\lim_{x \rightarrow c}$ exists, prove that $\lim_{x \rightarrow c} \sqrt{f} = \sqrt{\lim_{x \rightarrow c} f}$.

Solution:

Chapter 3: Some Extensions of the Limit Concept

Exercise 4.3.4

Let $c \in \mathbb{R}$ and let f be defined for $x \in (c, \infty)$ and $f(x) > 0$ for all $x \in (c, \infty)$. Show that $\lim_{x \rightarrow c} f = \infty$ if and only if $\lim_{x \rightarrow c} \frac{1}{f} = 0$.

Solution:

Exercise 4.3.8

Let f be defined on $(0, \infty)$ to $\mathbb{R}$. Prove that $\lim_{x \rightarrow \infty} f = L$ if and only if $\lim_{x \rightarrow 0^+} f\left(\frac{1}{x}\right) = L$.

Solution:

Exercise 4.3.12

Find functions f and g defined on $(0, \infty)$ such that $\lim_{x \rightarrow \infty} f = \infty$ and $\lim_{x \rightarrow \infty} g = \infty$ and $\lim_{x \rightarrow \infty} (f - g) = 0$. Can you find such functions, with $g(x) > 0$ for all $x \in (0, \infty)$ such that $\lim_{x \rightarrow \infty} \frac{f}{g} = 0$?

Solution:

Exercise 4.3.13

Let f and g be defined on (a, ∞) and suppose that $\lim_{x \rightarrow \infty} f = L$ and $\lim_{x \rightarrow \infty} g = \infty$. Prove that $\lim_{x \rightarrow \infty} f \circ g = L$.